

AT&T Global Information Solutions Servers

This information is summary in nature and supplements your other sales materials. Aberdeen Group believes it to be accurate as of the date of issue. The analysis contained in this report represents the opinion of Aberdeen Group based on our review of publicly-available material. This report is IBM confidential.

AT&T Global Information Solutions Server Product Overview

- ◆ **Desktop to massively parallel product line**
- ◆ **All servers based on Intel 486 and Pentium microprocessor. No RISC servers.**
- ◆ **Unix System V.4 MP OS with Reliability, Availability, Serviceability enhancements**
- ◆ **Low-end servers also run OS/2, SCO Unix, NetWare, and Windows NT**
- ◆ **Microchannel peripheral bus adapters**
- ◆ **Full line of peripherals, including RAID storage**
- ◆ **High availability clustering with LifeKeeper is comparable to HACMP**

AT&T Global Information Solutions Server Products

◆ Model 3555 midrange server

- Available May 31, 1994
- With 4 Pentium processors, 1,296 TPM-C at \$976 per tpmC.
\$1.265 million TPC-C life cycle cost
- Base 4-way Pentium CEC is \$120,600 U.S. list

AT&T System 3000 Servers

Model	Processor/ Quantity	Processor Type	Cache per CPU	Memory Range	Slots	Internal Full Height DASD
3450/x	1-4	Intel 486	256KB	16MB - 1 GB	7	5
3455/x	1-6	Pentium	256 KB	16 MB - 1 GB	7	5
3520/x	1-8	Intel 486	256 KB	32 MB - 2 GB	8 or 16	31
3525/x	1-8	Pentium	256 KB	32 MB - 2 GB	8 or 16	31
3550/x	2,4,6,8	Intel 486	512 KB	32 MB - 2 GB	16	31
3555/x	2,4,6,8	Pentium	256 KB + 8 MB	32 MB - 2 GB	16	31

AT&T Global Information Solutions

Prioritized Sales Messages

- ◆ NCR is now Global Information Solutions, the computer arm of AT&T
- ◆ GIS is focused on *Customer Information Solutions* -- “IT solutions that help an organization better know its customers and prospects by more effectively getting, moving, and using information about them”
- ◆ AT&T GIS “combines the best in computing and communications”
- ◆ Seeks market leadership in key industries: retail, financial services, computer product manufacturing, communications, transportation, and government markets
- ◆ Focus on *solutions* that improve customer’s customer relationships and increase customer’s revenues, not just on cost cutting

AT&T Global Information Solutions

Prioritized Sales Messages (Cont.)

- ◆ **The market leader in Intel-based open systems computing. Often a winner for large distributed/replicated bids (i.e., 4,000 Walmart stores).**
- ◆ **Offers excellent price/performance (see excellent TPC-A, TPC-C), value, service**
- ◆ **Industry leader in complex decision support with Teradata**
- ◆ **Specialized solutions and greatest market share in retail and banking industries. Beginning to build on AT&T products/name in telecommunications**
- ◆ **Committed to leadership in the emerging market for Windows NT servers**

AT&T GIS Strengths

- ◆ **Breadth of compatible server products is excellent**
- ◆ **Good, often leading, TPC price/performance**
- ◆ **The backing of AT&T**
- ◆ **Bullet-proof version of UNIX. Known for stability.**
- ◆ **The Intel multiprocessor market champion (vs. RISC)**
- ◆ **Market strength and broad solutions strength in retail and banking**

AT&T Global Information Solutions Customer Concerns

- ◆ **Acquisition of NCR by AT&T in 1991 has not jelled. Numerous management and strategy changes. Not profitable of late.**
- ◆ **High-end model 3600 cluster-in-a-box has not proven performance potential. Bottlenecks? Where is the 3700 replacement for the 3600?**
- ◆ **High-end Teradata DBC 1012 was decision-support specialist. Expensive. Not very fast. At end of life. Market is now looking to new-breed solutions (e.g., IBM SP2)**
- ◆ **Sales force quality varies from excellent to spotty. Recent reorg into 500 worldwide key account teams. Will the sales force work hard to support customers?**

AT&T Global Information Solutions

Customer Concerns (Cont.)

- ◆ List prices seem high [but market prices/quotes are very competitive].
- ◆ NCR name was seldom known in the datacenter (outside retail and banking). AT&T GIS is basically a new *potential* supplier to broad market.
- ◆ Poor channel program and late efforts in partnering mean fewer (and later arriving) application solutions and solution providers
- ◆ Sybase Data Navigator MPP RDBMS program delayed, revised. Relationship is not strong.

AT&T Global Information Solutions Profile

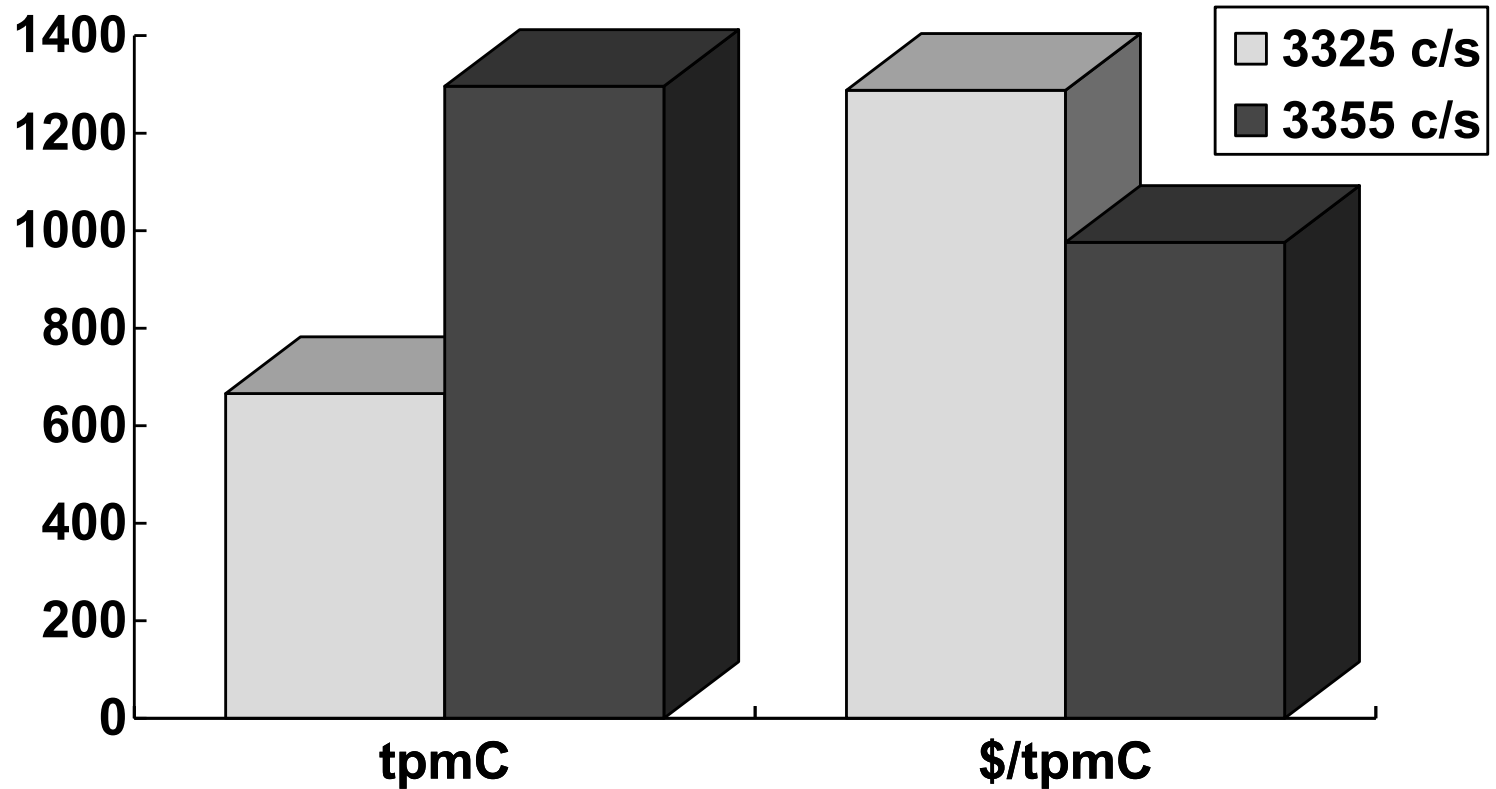
◆ The strength of AT&T

- World's largest technology company
- World's second largest company (market capitalization)
- \$68 billion in 1993 revenues -- GIS about \$7 billion
- 312,700 employees -- 20,000 are in GIS
- 8,560 locations in 120 countries

◆ In the News

- Pentium model refresh
- NCR becomes AT&T Global Information Solutions
- AT&T GIS will port LAN Manager for UNIX to AIX and co-market with IBM

AT&T Global Information Solutions



Data General Servers

This information is summary in nature and supplements your other sales materials. Aberdeen Group believes it to be accurate as of the date of issue. The analysis contained in this report represents the opinion of Aberdeen Group based on our review of publicly-available material. This report is IBM confidential.

Data General Server Product Overview

◆ Aviion

- **Deskside to 16-processor symmetric multiprocessing**
- **All servers based on Motorola 88K RISC microprocessors**
- **High-availability clustering up to 4 systems**
- **RAID 0, 1, 3, 5, up to 2 TB external storage**
- **Sold as commercial server, strong in medical market**
- **Aggressive pricing**

Data General Aviion Servers

Model	Processor/ Quantity	Processor Type	Cache per CPU	Memory Range	Slots (Internal/ External)	Internal Full Height DASD
4500	1	88110	16 KB	16 MB - 256 MB	NA/0	2 GB
5500	1-2	88110	256 KB	32 MB - 256 MB	2/0	8 GB
8500	2-4	88110	512 KB	64 MB - 2 GB	5/0	10 GB
9500	2-16	88110	1 MB	128 MB - 2 GB	5/16	10 GB

Data General Server Products

◆ **Aviion 4500, 5500, 8500, 9500**

- **Low-end branch-office to 16-CPU SMP**
- **DG/UX is UNIX SVR4**
- **2 TB max. external storage for 9500**
- **Offers Oracle Parallel Server 2-4-system clustering**
- **RAID 0, 1, 3, 5; SCSI-2 drives**
- **CA-UNICENTER marketing/service and support**
- **Entry price \$14,490 (5500), \$40,995 (8500), \$79,995 (9500)**
 - » **5500 130.9 tps-A, \$5,780 \$/tps-A**
 - » **9500 523.64 tps-A, \$5,357 tps-A**

Data General Prioritized Sales Messages

- ◆ **Open systems**
- ◆ **High availability (RAID for disks, tape drives)**
- ◆ **Price-performance**
- ◆ **Alternate choice for MIS shops wanting to downsize the data center**

Data General Strengths/Concerns

◆ Strengths:

- **Technology:** early into SMP, RAID; extended security; NetWare support; LAN/WAN networking (Token Ring, Netbios, NetWare, SNA, DECnet, LAN Manager)
- **Aggressive pricing/price-performance**

◆ Customer Concerns:

- **DG financial problems**
- **Future of chip choice (Motorola 88K chipset end of life)**

Data General Profile

- ◆ **Annual Sales: \$1 billion fiscal '93**
- ◆ **Strong in medical market via VARs**
- ◆ **Focus on government, health-care, finance, retail, and telecomm. markets**
- ◆ **50/50 international/US sales; strictly commercial server (no technical/workstation sales)**
- ◆ **Distribution: 50/50 direct/VAR**
- ◆ **In The News**
 - loss in 1Q94

Digital Equipment Corporation Servers

This information is summary in nature and supplements your sales materials. Aberdeen Group believes it to be accurate as of the date of issue. The analysis contained in this report represents the opinion of Aberdeen Group based upon our review of publicly available material. This report is IBM Confidential.

Digital Server Products

<u>Server Model</u>	<u>Entry Price</u>	<u>Fully Configured</u>
DEC 2000/500 AXP	\$10,600	
DEC 3000/600S AXP	\$21,190	
DEC 3000/800S AXP	\$34,130	
DEC 4000/700 AXP	\$43,000	\$87,160
DEC 2100/A500MP	\$18,900	\$36,900
DEC 7000/600 AXP	\$126,334	\$149,000
DEC 2000/300 AXP	\$7,799	\$9,499
DEC 10000/600AXP	\$210,634	\$225,300

Digital Alpha AXP Servers

Model	Processor/ Quantity	Processor Type	Cache per CPU	Memory Range	Slots
2000/300 AXP	1	Alpha AXP	256 KB	32 MB - 256 MB	6
2000/500 AXP	1	Alpha AXP	256 KB	32 MB - 256 MB	6
3000/600S AXP	1	Alpha AXP	2 MB	64 MB - 512 MB	6
3000/800S AXP	1	Alpha AXP	2 MB	64 MB - 1 GB	6
4000/700 AXP	1-2	Alpha AXP	1 MB	64 MB - 2 GB	6
2100/A500MP	1-4	Alpha AXP	8 KB	64 MB - 2 GB	11
7000/600 AXP	1-6	Alpha AXP	4 MB	128 MB - 14 GB	48
10000/600 AXP	1-6	Alpha AXP	4 MB	128 KB - 14 GB	48

Digital Prioritized Sales Messages

- ◆ **Alpha performance combined with aggressive pricing has created the best price/performance in the industry.**
- ◆ **DEC is creating software frameworks for enterprise-wide open computing to meet customer needs for applications interoperability from the desktop to datacenter.**
- ◆ **DEC is committed to open systems properly implemented to provide customers with real benefits in a production environment.**
- ◆ **DEC is partnering with Microsoft to provide users with the most powerful implementation of NT Advanced Server on Alpha. Combined with Digital's network management and communications enhancements this platform will be the one departmental and workgroup computing platform of the future.**
- ◆ **DEC is the first and only company to implement the OSF/1 operating system. DEC's OSF/1 meets the specifications of more standards setting groups than any other Unix operating system.**
- ◆ **DEC's worldwide professional services organization has the skills and capabilities to meet the needs of the largest global enterprises that are implementing open systems consisting of components from diverse suppliers and competitors.**

Digital Equipment Corporation Strengths

- ◆ **Digital's worldwide multi-vendor field service organization.**
- ◆ **Digital's acknowledged expertise in putting together large, complex computer networks.**
- ◆ **The internal clock rate of Digital's Alpha AXP microprocessor.**

Digital Equipment Corporation Concerns

- ◆ **DEC's share of the multiuser commercial Unix market is insignificant. With the current neutral attitude of prospects to DEC's limited number of Alpha OSF/1 DECsystems, DEC is unlikely to establish a critical mass for long-term viability.**
- ◆ **DEC is in the midst of yet another retrenching. As a result, prospects are concerned about which customer support people may be let go and which promised products will be eliminated.**
- ◆ **DEC is not yet able or willing to release TPC-C benchmark results. There is a growing misgiving that Alpha systems are not as fast as DEC's internal processor results indicated.**
- ◆ **DEC is not a major player in Unix - the bulk of its revenues are still derived from VMS - while most multiuser independent software vendors are delivering their applications first on Unix.**
- ◆ **At the beginning of 1993, DEC promised Microsoft NT servers and personal systems at Intel PC prices by the third quarter of 1993 - and they are still not announced.**
- ◆ **As DEC goes through the transition from 32 bit VAX/VMS to 64 bit Alpha/Open VMS systems, the installed base is very concerned about software conversions and transition glitches.**
- ◆ **Alpha is a 64 bit processor. But, until DEC redoes its middleware to support 64 bits, its users will not obtain the full performance advantages promised.**

Digital Equipment Corporation Profile

◆ **Annual Sales: \$14.4 billion in FY93**

◆ **Number of employees: 94,200**

◆ **Distribution:**

- **68% Direct**
- **28% VARs and Distributors**
- **4% OEM**

◆ **Target Markets:**

- **Aerospace**
- **Banking**
- **Education/Research**
- **Electronic**
- **Government/Military**

Digital Equipment Corporation Profile

◆ Target Markets (continued):

- Healthcare
- Insurance
- Manufacturing
- Oil/Gas
- Automotive
- Architecture/Engineering/Construction
- Pharmaceutical
- Publishing
- Retail
- Securities
- Telecommunications
- Transportation
- Utilities

Digital Equipment Corporation Workstations

This information is summary in nature and supplements your sales materials. Aberdeen Group believes it to be accurate as of the date of issue. The analysis contained in this report represents the opinion of Aberdeen Group based upon our review of publicly available material. This report is IBM Confidential.

Digital Workstation Products

<u>Workstation Model</u>	<u>Entry Price</u>	<u>Fully Configured</u>
DEC 3000/300X AXP	\$5,295	
DEC 3000/300LX AXP	\$10,495	
DEC 3000/600 AXP	\$19,995	
DEC 3000/800 AXP	\$36,000	

Digital Alpha AXP Workstations

Model	Processor/ Quantity	Processor Type	Cache per CPU	Memory Range	Slots
3000/300X AXP	1	Alpha AXP	256 KB	32 MB - 256 MB	6
3000/300LX AXP	1	Alpha AXP	256 KB	32 MB - 256 MB	6
3000/600 AXP	1	Alpha AXP	2 MB	64 MB - 512 MB	6
3000/800 AXP	1	Alpha AXP	2 MB	64 MB - 1 GB	6

Digital Prioritized Sales Messages

- ◆ Alpha performance combined with aggressive pricing has created the best price/performance in the industry.
- ◆ DEC is creating software frameworks for enterprise-wide open computing to meet customer needs for applications interoperability from the desktop to datacenter.
- ◆ DEC is committed to open systems properly implemented to provide customers with real benefits in a production environment.
- ◆ DEC is the first and only company to implement the OSF/1 operating system. DEC's OSF/1 meets the specifications of more standards setting groups than any other Unix operating system.
- ◆ DEC's worldwide professional services organization has the skills and capabilities to meet the needs of the largest global enterprises that are implementing open systems consisting of components from diverse suppliers and competitors.

Digital Equipment Corporation Strengths

- ◆ Digital's worldwide multi-vendor field service organization.
- ◆ Digital's acknowledged expertise in putting together large, complex computer networks.
- ◆ The internal clock rate of Digital's Alpha AXP microprocessor.

Digital Equipment Corporation Concerns

- ◆ DEC's history in the workstation marketplace has been to introduce products and then withdraw them. With DEC's share of the workstation marketplace very small, its products considered "okay, but not exceptional", minimal ISV support, and a need to reduce costs, the DECstation product line could possibly be dropped.
- ◆ DEC is in the midst of yet another retrenching. As a result, prospects are concerned about which customer support people may be let go and which promised products will be eliminated.
- ◆ DEC is not yet able or willing to release TPC-C benchmark results. There is a growing misgiving that Alpha systems are not as fast as DEC's internal processor results indicated.
- ◆ DEC is not a major player in Unix - the bulk of its revenues are still derived from VMS - while most multiuser independent software vendors are delivering their applications first on Unix.
- ◆ At the beginning of 1993, DEC promised Microsoft NT servers and personal systems at Intel PC prices by the third quarter of 1993 - and they are still not announced.

Digital Equipment Corporation Concerns

- ◆ As DEC goes through the transition from 32 bit VAX/VMS to 64 bit Alpha/Open VMS systems, the installed base is very concerned about software conversions and transition glitches.
- ◆ Alpha is a 64 bit processor. But, until DEC redoes its middleware to support 64 bits, its users will not obtain the full performance advantages promised.

Digital Equipment Corporation Profile

◆ **Annual Sales: \$14.4 billion in FY93**

◆ **Number of employees: 94,200**

◆ **Distribution:**

- **68% Direct**
- **28% VARs and Distributors**
- **4% OEM**

◆ **Target Markets:**

- **Aerospace**
- **Banking**
- **Education/Research**
- **Electronic**
- **Government/Military**

Digital Equipment Corporation Profile

◆ Target Markets (continued):

- Healthcare
- Insurance
- Manufacturing
- Oil/Gas
- Automotive
- Architecture/Engineering/Construction
- Pharmaceutical
- Publishing
- Retail
- Securities
- Telecommunications
- Transportation
- Utilities

Hewlett-Packard Servers

This information is summary in nature and supplements your sales materials. Aberdeen Group believes it to be accurate as of the date of issue. The analysis contained in this report represents the opinion of Aberdeen Group based upon our review of publicly-available material. This report is IBM Confidential.

HP Server Products

<u>Server Model</u>	<u>Entry Price</u>	<u>Fully Configured</u>
HP 9000/800E	\$4,130	\$24,840
HP 9000/800F	\$8,700	\$35,565
HP 9000/800G	\$15,000	\$93,440
HP 9000/800H	\$24,000	\$121,580
HP 9000/800I	\$58,500	\$137,580
HP 9000/T500	\$165,000	\$544,200
HP 9000/890	\$120,000	\$534,000

HP 9000 Series 800 Servers

Model	Processor/ Quantity	Processor Type	Cache per CPU	Memory Range	Slots	Internal Full Height DASD
800E	1	PA-RISC	64-256 KB	16 MB - 512 MB	2-4	2
800G	1	PA-RISC	256KB - 2 MB	32 MB - 768 MB	4	2
800H	1	PA-RISC	64 KB - 2 MB	64 MB - 768 MB	8	3
800I	1	PA-RISC	64 KB - 2 MB	64 MB - 768 MB	12	3
T500	1-12	PA-RISC	1 MB	256MB - 2 GB	14-112	NA

HP Prioritized Sales Messages

- ◆ **HP's high performance and proven scalable PA-RISC architecture provides the underlying technological advantage over its competitors in the Unix workstation and server markets.**
- ◆ **HP is #1 in terms of revenue in the commercial RISC/Unix server market and #2 in the Unix workstation market (Source: Aberdeen Group, Inc.). In addition, HP's Unix server business grew at over 35% in 1993 and its workstation business at over 20%. HP continues to gain market share.**
- ◆ **As the world's second largest computer company, HP is profitable and has been in business for over 50 years with stable executive management. Fortune magazine and other publications have honored HP as the most admired company in the computer industry.**
- ◆ **HP is a leader in industry standards setting bodies and freely licenses its technology for establishing standards. Examples include HP's technological contributions to the Open Software Foundation (OSF), Object Management Group (OMG), Committee for Open Systems (COSE), and the Precision RISC Organization (PRO).**
- ◆ **HP is committed to enhancing the middleware that runs on HP-UX to meet industry standards and provide its customers with the most reliable, available, and serviceable Unix operating environment in the industry.**

HP Prioritized Sales Messages

- ◆ **HP has the best relationships with the greatest number of independent software vendors whose products customers want. HP is consistently rated by VAR surveys as having the best VAR program in the industry.**
- ◆ **HP's Corporate Business Server T500 provides the highest proven throughput in the industry from an open Unix server @ 2,110 TPC-C transactions per minute.**
- ◆ **HP offers the most choices of servers in the industry through slot expansions and performance.**
- ◆ **Servers are compatible and upgradeable from the under \$10,000 E25 to the datacenter class 12-way T500.**

HP Strengths

- ◆ **PA-RISC architecture provides HP with a road map to the future.**
- ◆ **HP provides the key platform for ISVs to port their products to as they move to Unix from mainframe and other proprietary platforms.**
- ◆ **HP's platform is the primary one for all of the major Unix DBMS suppliers such as Oracle, Informix, Sybase, Ingres, Software AG, etc.**
- ◆ **HP's high-end SMP implementation provides the highest throughput in the industry.**
- ◆ **HP has its own worldwide direct sales force and field service organization.**

HP Concerns

- ◆ **HP will not provide sales coverage to non-targeted accounts - no matter how large and how well budgeted their open systems project.**
- ◆ **HP servers require 25 minutes to recover from a systems crash, even if caused by a software error.**
- ◆ **HP field sales and professional services organization appears to be of lower quality and competency than its competitors.**
- ◆ **HP's order administration system makes it very difficult to obtain firm quotes and delivery dates. Maintenance and service contracts are difficult to administer and ordered product deliveries are relatively haphazard.**

HP Profile

- ◆ **Annual Sales: \$20.3 billion in FY93**
- ◆ **Number of employees: 96,200**
- ◆ **Distribution:**
 - **75% Direct**
 - **25% VARs and Distributors**
 - **0% OEM**
- ◆ **Target Markets:**
 - **Aerospace**
 - **Banking**
 - **Education/Research**
 - **Electronic**
 - **Government/Military**

HP Profile (Cont.)

◆ Target Markets (continued):

- Healthcare
- Insurance
- Manufacturing
- Oil/Gas
- Automotive
- Architecture/Engineering/Construction
- Pharmaceutical
- Publishing
- Retail
- Securities
- Telecommunications
- Transportation
- Utilities

Hewlett-Packard Workstations

This information is summary in nature and supplements your sales materials. Aberdeen Group believes it to be accurate as of the date of issue. The analysis contained in this report represents the opinion of Aberdeen Group based upon our review of publicly available material. This report is IBM Confidential.

HP Workstation Products

<u>Workstation Model</u>	<u>Entry Price</u>	<u>Fully Configured</u>
Model 712/60	\$3,995	\$13,350
Model 712/80	\$9,320	\$16,900
Model 715/64	\$9,995	\$13,295
Model 715/80	\$13,600	\$16,900
Model 715/100	\$19,005	\$22,305
Model 735/125	\$37,295	\$51,145
Model 715/33	\$6,200	\$16,395
Model 715/50	\$9,030	\$25,125
Model 725/50	\$13,010	\$29,905
Model 715/75	\$13,010	\$22,605
Model 725/75	\$16,010	\$25,605
Model 735	\$28,300	\$41,865
Model 755	\$40,995	\$49,995

HP 9000 Series 700 Workstations

Model	Processor/ Quantity	Processor Type	Cache per CPU	Memory Range	Slots	Internal Full Height DASD
712	1	PA-RISC	64 KB-256 KB	16 MB - 128 MB	2	1
715	1	PA-RISC	64 KB-256 KB	16 MB - 256 MB	1/5	2
725	1	PA-RISC	256 KB	32 MB - 256 MB	5	2
735	1	PA-RISC	256 KB	32 MB - 400 MB	1	2
755	1	PA-RISC	256 KB	64 MB - 768 MB	4	2

HP Prioritized Sales Messages

- ◆ **HP's high performance and proven scalable PA-RISC architecture provides the underlying technological advantage over its competitors in the Unix workstation and server markets.**
- ◆ **HP is #1 in terms of revenue in the commercial RISC/Unix server market and #2 in the Unix workstation market (Source: Aberdeen Group, Inc.). In addition, HP's Unix server business grew at over 35% in 1993 and its workstation business at over 20%. HP continues to gain market share.**
- ◆ **As the world's second largest computer company, HP is profitable and has been in business for over 50 years with stable executive management. Fortune magazine and other publications have honored HP as the most admired company in the computer industry.**
- ◆ **HP is a leader in industry standards setting bodies and freely licenses its technology for establishing standards. Examples include HP's technological contributions to the Open Software Foundation (OSF), Object Management Group (OMG), Committee for Open Systems (COSE), and the Precision RISC Organization (PRO).**
- ◆ **HP is committed to enhancing the middleware that runs on HP-UX to meet industry standards and provide its customers with the most reliable, available, and serviceable Unix operating environment in the industry.**

HP Prioritized Sales Messages (Cont.)

- ◆ **HP has the best relationships with the greatest number of independent software vendors whose products customers want. HP is consistently rated by VAR surveys as having the best VAR program in the industry.**
- ◆ **The best kept secret in the industry: HP workstation graphics are the fastest in the industry- faster than even Silicon Graphics.**
- ◆ **HP's 700 Series offers the best price/performance starting at under \$4,000 for a complete system and maintains this with the fastest workstations available in the industry.**

Hewlett-Packard Co. Strengths

- ◆ **PA-RISC architecture provides HP with a road map to the future.**
- ◆ **HP provides the key platform for ISVs to port their products to as they move to Unix from mainframe and other proprietary platforms.**
- ◆ **HP's platform is the primary one for all of the major Unix DBMS suppliers such as Oracle, Informix, Sybase, Ingres, Software AG, etc.**
- ◆ **HP's high-end SMP implementation provides the highest throughput in the industry.**
- ◆ **The workstation environment of HP has the best combination of high speed graphic and floating point performance.**
- ◆ **HP has its own worldwide direct sales force and field service organization.**

Hewlett-Packard Co. Concerns

- ◆ HP will not provide sales coverage to non-targeted accounts - no matter how large and how well budgeted their open systems project.
- ◆ HP servers require 25 minutes to recover from a systems crash, even if caused by a software error.
- ◆ HP field sales and professional services organization appears to be of lower quality and competency than its competitors.
- ◆ HP's order administration system makes it very difficult to obtain firm quotes and delivery dates. Maintenance and service contracts are difficult to administer and ordered product deliveries are relatively haphazard.

Hewlett-Packard Co. Profile

- ◆ **Annual Sales: \$20.3 billion in FY93**
- ◆ **Number of employees: 96,200**
- ◆ **Distribution:**
 - **75% Direct**
 - **25% VARs and Distributors**
 - **0% OEM**
- ◆ **Target Markets:**
 - **Aerospace**
 - **Banking**
 - **Education/Research**
 - **Electronic**
 - **Government/Military**

Hewlett-Packard Co. Profile

◆ Target Markets (continued):

- Healthcare
- Insurance
- Manufacturing
- Oil/Gas
- Automotive
- Architecture/Engineering/Construction
- Pharmaceutical
- Publishing
- Retail
- Securities
- Telecommunications
- Transportation
- Utilities

Sequent Servers

This information is summary in nature and supplements your other sales materials. Aberdeen Group believes it to be accurate as of the date of issue. The analysis contained in this report represents the opinion of Aberdeen Group based on our review of publicly-available material. This report is IBM confidential.

Sequent Server Product Overview

◆ Symmetry, WinServer

- Deskside to 30-processor symmetric multiprocessing
- All servers 486/Pentium-based; no RISC
- Low-end WinServer runs Windows NT
- High-availability Sequent Clusters up to 4 systems
- RAID 0, 1, 2
- DYNIX/ptx OS just moved up to UNIX V.4
- 10K packet/sec Ethernet, 240MB/sec VMEbus on new 5000s
- 840GB SCSI-2 disks

Sequent Symmetry Servers

Model	Processor/ Quantity	Processor Type	Cache per CPU	Memory Range	Slots	Internal Full Height DASD
ELS	1	Intel 486	256 KB	96 MB - 384 MB	6/8	2/9
2000/290	2-6	Pentium/60	1 MB	64 MB - 320 MB	5	5
2000/490	2-10	Pentium/60	1 MB	64 MB - 768 MB	9	4
2000/790	2-30	Pentium/60	1 MB	64 MB - 1.6 GB	9	12
5000/SE20	2-10	Pentium/66	1 MB	64 MB - 2 GB	9	7
5000/SE60	2-30	Pentium/66	1 MB	64 MB - 2 GB	9	12/24

Sequent Server Products

◆ Symmetry ELS, 2000, 5000

- ELS Desktop, Deskside, II Deskside
- 2000 midrange
 - » Entry price \$85,000 (290), \$155,000 (490), \$351,000 (790)
 - » Benchmarks 2000/250 183.34 tps-A, 2000/750 2-cluster 1002.37 tps-A
- New 5000 high-end
 - » Entry price \$171,100 (SE20), \$469,800 (SE60)
 - » Fully configured \$1,030,400 (SE20), \$2,625,700 (SE60)
 - » No benchmarks yet for 5000

◆ WinServer 500, 1000, 1500, 3000, 5000

- Windows NT
- 500 deskside (1-2 CPUs), 1000/1500 up to 50 fulltime users (1-2 CPUs), 3000 hundreds of users (1-6 CPUs), 5000 high-end (2-30 CPUs)
- Entry prices \$13,200 (500), \$29,900 (1000), \$39,300 (1500), \$79,100 (3000), \$246,900 (5000)

Sequent Sales Messages

- ◆ **Open systems (early into high-end Unix)**
- ◆ **Open-system scalability (SMP, clustering, RDBMS parallelism, broad product/performance range)**
- ◆ **Development relationships with Oracle, Informix (parallelism 6-month exclusive on Informix DSA)**
- ◆ **Best-in-class SMP parallelism (up to 30 SMP processors)**
- ◆ **Windows NT on Intel for early adopters**

Sequent Strengths/Concerns

◆ Strengths:

- High-end Unix-system market share
- SMP plus clustering scalability
- Good RDBMS performance
- Entrepreneurial. Good at closing prospects

◆ Customer Concerns:

- High pricing . . . mediocre price/performance
- No focus on low-end server, workstation
- Fewer support resources/experience
- Under-\$500-million size
- Financial losses. Lack of long-term profitability

Sequent Profile

- ◆ **Annual Sales: \$354 million fiscal '93**
- ◆ **Distribution: direct**
- ◆ **In The News**
 - **New service division with large number of consultant hires**
 - **Symmetry 5000**

Sun Microsystems, Inc. Servers

This information is summary in nature and supplements your sales materials. Aberdeen Group believes it to be accurate as of the date of issue. The analysis contained in this report represents the opinion of Aberdeen Group based upon our review of publicly available material. This report is IBM Confidential.

Sun Server Products

<u>Server Model</u>	<u>Entry Price</u>	<u>Fully Configured</u>
SPARCserver 1000	\$14,500	\$30,000
SPARCserver 2000	\$52,700	\$59,000

Sun SPARCserver Servers

Model	Processor/ Quantity	Processor Type	Cache per CPU	Memory Range	Slots	Internal Full Height DASD
SPARCserver 1000	1-8	SuperSPARC	1 MB	32 MB - 2 GB	3-12	4
SPARCserver 2000	2-20	SuperSPARC	1 MB	64 MB - 5 GB	4-40	18

Sun Prioritized Sales Messages

- ◆ **Originator and still leader in bringing the advantages of open systems to the marketplace.**
- ◆ **Sun is the client-server computing company - the only hardware supplier exclusively offering a common desktop and multiuser server hardware architecture and operating environment. Sun is the innovator in providing networked client/server solutions to the industry.**
- ◆ **Sun maintains both the lowest entry price and lowest price/performance point for commercial servers.**
- ◆ **Number 1 in Unix systems shipments and installed systems according to industry analysts.**
- ◆ **Leader in symmetrical multiprocessing (SMP) from the desktop to the datacenter. SMP is a requirement to meet users' hard-to-project, but rapidly increasing, throughput requirements.**
- ◆ **A \$4 billion worldwide corporation focused on delivering open systems to the marketplace.**

Sun Prioritized Sales Messages (Cont.)

- ◆ **Solaris 2.3, Sun's new Unix operating system, was developed specifically to provide commercial users with greater than mainframe reliability, availability, and security.**
- ◆ **Can use workstation servers for PC print, file, and maintenance over the life of the systems.**
- ◆ **If Sun's prices ever become too high there is an army of third party suppliers to whom Sun has licensed the SPARC architecture and Solaris operating system eagerly waiting to enter the marketplace.**
- ◆ **Sun's positive working relationships with value-added resellers and third party maintenance support providers assures lowest cost of acquisition and maintenance over the life of the systems.**
- ◆ **The leading partner with independent software vendors (ISVs). Most ISVs use Sun systems to develop their client/server applications assuring Sun customers that they will receive the latest releases of software first and it will be the most bug free and best performing.**

Sun Strengths

- ◆ Large number of applications.
- ◆ NFS file serving.
- ◆ Champion of the “common” customer.
- ◆ Will sell through any distributor. (You never know where your Sun competition is coming from.)
- ◆ Unix application development programmer’s workbench.

Sun Concerns

- ◆ Sun field support is limited and, in many cases, contracted out to other firms.
- ◆ There are very few references for running SPARCcenter 2000s and SPARCserver 1000s for business critical production OLTP applications.
- ◆ User's transition from SunOS, Sun's previous Unix derivative operating system, to Solaris 2.3 has been very difficult and Sun has provided minimal technical support from its corporate offices.
- ◆ When SPARCservers crash, data can be lost and it can take up to 15 hours to rebuild the system - including reinstalling the operating system.
- ◆ Sun's commercial client/server strategy is highly dependent on implementing Sybase front-end GUIs on a SPARCstation and server software on a SPARCserver. So far, very few application developers have been able to accomplish this.
- ◆ Real world scalability of applications appears to be very poor on all of Sun's workstations and servers.
- ◆ Sun does not appear to be a major player in the new generation of standards setting groups. Many Sun products, such as its critical OpenLook GUI, may be excluded from future standards.

Sun Profile

- ◆ **Annual Sales: \$4.3 billion in FY93**
- ◆ **Number of employees: 13,253**
- ◆ **Distribution:**
 - **48% Direct**
 - **37% VARs and Distributors**
 - **15% OEM**
- ◆ **Target Markets:**
 - **Aerospace**
 - **Banking**
 - **Education/Research**
 - **Electronic**
 - **Government/Military**

Sun Profile (Cont.)

◆ Target Markets (continued):

- Healthcare
- Insurance
- Manufacturing
- Oil/Gas
- Automotive
- Architecture/Engineering/Construction
- Pharmaceutical
- Publishing
- Retail
- Securities
- Telecommunications
- Transportation
- Utilities

Sun Microsystems Workstations

This information is summary in nature and supplements your sales materials. Aberdeen Group believes it to be accurate as of the date of issue. The analysis contained in this report represents the opinion of Aberdeen Group based upon our review of publicly available material. This report is IBM Confidential.

Sun Workstation Products

<u>Workstation Model</u>	<u>Entry Price</u>	<u>Fully Configured</u>
Classic	\$1,895	\$6,095
SPARCstation 5-70	\$3,995	\$10,395
SPARCstation 5-85	\$11,395	\$19,995
SPARCstation 10-40	\$14,495	\$37,745
SPARCstation 10-51	\$18,495	\$44,745
SPARCstation 10-402MP	\$17,485	\$34,995
SPARCstation 10-512MP	\$23,995	\$34,995
SPARCstation 10-514MP	\$39,995	\$41,995
SPARCstation 20-50	\$8,650	\$9,600
SPARCstation 20-51	\$14,195	\$24,995
SPARCstation 20-61	\$12,850	\$26,995
SPARCstation 20-502MP	\$14,195	\$24,995
SPARCstation 20-612MP	\$22,495	\$31,495
SPARCstation 20-514MP	\$29,995	\$32,995

Sun SPARCstation Workstations

Model	Processor/ Quantity	Processor Type	Cache per CPU	Memory Range	Slots	Internal Full Height DASD
5-70, -85	1	microSPARC II	16 KB	16 MB - 256 MB	3	2
10-40, 10-51	1	SuperSPARC	36 KB	32 MB - 512 MB	4	2
10-402, -512, -514 MP	2,2,4	SuperSPARC	36 KB	32 MB - 512 MB	4	2
20-50, -51, -61	1	SuperSPARC	36 KB	32 MB - 512 MB	4	2
20-502, -612, -514 MP	2,2,4	SuperSPARC	36 KB	32 MB - 512 MB	4	2

Sun Prioritized Sales Messages

- ◆ **Originator and still leader in bringing the advantages of open systems to the marketplace.**
- ◆ **Always has and always will have the lowest entry price for powerful Unix workstations on the desktop.**
- ◆ **Number 1 in Unix systems shipments and installed systems according to industry analysts.**
- ◆ **With Sun's WABI, you can have a RISC/Unix workstation while being able to run leading MS Windows application binaries.**
- ◆ **A \$4 billion worldwide corporation focused on delivering open systems to the marketplace.**

Sun Prioritized Sales Messages (Cont.)

- ◆ **If Sun's prices ever become too high there is an army of third party suppliers to whom Sun has licensed the SPARC architecture and Solaris operating system eagerly waiting to enter the marketplace.**
- ◆ **Sun's positive working relationships with value-added resellers and third party maintenance support providers assures lowest cost of acquisition and maintenance over the life of the systems.**
- ◆ **The leading partner with independent software vendors (ISVs). Most ISVs use Sun systems to develop their client/server applications assuring Sun customers that they will receive the latest releases of software first and it will be the most bug free and best performing.**

Sun Strengths

- ◆ **Low cost provider of workstations.**
- ◆ **Large number of applications**
- ◆ **Champion of the “common” customer.**
- ◆ **Will sell through any distributor. (You never know where your Sun competition is coming from.)**
- ◆ **Unix application development programmer’s workbench.**

Sun Concerns

- ◆ Sun field support is limited and, in many cases, contracted out to other firms.
- ◆ SPARCstations provide the least floating point performance in the industry and have the lowest floating point price/performance.
- ◆ User's transition from SunOS, Sun's previous Unix derivative operating system, to Solaris 2.3 has been very difficult and Sun has provided minimal technical support from its corporate offices.
- ◆ Sun's WABI (Windows Application Binary Interface) that is supposed to support MS Windows-based applications running on SPARCstations, is certified for only a very few software packages and even on these applications it is considered buggy.
- ◆ Real world scalability of applications appears to be very poor on all of Sun's workstations and servers.
- ◆ Sun does not appear to be a major player in the new generation of standards setting groups. Many Sun products, such as its critical OpenLook GUI, may be excluded from future standards.

Sun Profile

- ◆ **Annual Sales: \$4.3 billion in FY93**
- ◆ **Number of employees: 13,253**
- ◆ **Distribution:**
 - **48% Direct**
 - **37% VARs and Distributors**
 - **15% OEM**
- ◆ **Target Markets:**
 - **Aerospace**
 - **Banking**
 - **Education/Research**
 - **Electronic**
 - **Government/Military**

Sun Profile (Cont.)

◆ Target Markets (continued):

- Healthcare
- Insurance
- Manufacturing
- Oil/Gas
- Automotive
- Architecture/Engineering/Construction
- Pharmaceutical
- Publishing
- Retail
- Securities
- Telecommunications
- Transportation
- Utilities